

Sirine SAHRAOUI

Electrical Engineer – Electronic & Hardware Design

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Profile

Electrical engineer specializing in Hardware Design, with experience in multilayer PCB design, hardware simulation, and validation. Proficient in Altium Designer, LTspice, and HyperLynx for schematic design, PCB layout, and SI/PI analysis. Solid knowledge of PCBA processes, PCB assembly, design for manufacturability (DFM), and PCB industrialization. Passionate about developing reliable hardware solutions for embedded and electronic systems.

Professional Experience

OTTO ES – Engineer in Electronic & Hardware Design

Nov 2025 – Mar 2026

- Design and industrialization of 8-layer boards, covering the full definition of manufacturing, assembly, and validation processes.
- Writing technical documentation and detailed work instructions for the production team.

Internships

Hardware Design Engineer Intern – OTTO ES

Feb – Aug 2025

- Designed a 6-layer PCB integrating a System-on-Module (SOM) with IoT modules (Wi-Fi, LTE, LoRa, GPS), communication, debugging, and power supply blocks.
- Contributed to industrialization and preparation of production schematics.
- Performed power integrity simulation with HyperLynx and validated voltage regulators via LTspice and PSpice for TI.

PCB Assembly Technician Intern – La Pratique Electronique (LPE)

Jul – Aug 2024

- Monitored the full PCBA manufacturing process from component preparation to final inspection, gaining hands-on experience in SMT and THT assembly.

Introductory Intern – Tunisie Telecom

Jul – Aug 2023

- Gained practical exposure to broadband access technologies and multiple access techniques (TDMA, FDMA).

Education

National Engineering School of Sfax (ENIS) – Electrical Engineering

2022–2025

Preparatory Institute for Engineering Studies of Nabeul (IPEIN) – Physics & Technology

2020–2022

Baccalaureate in Science & Technology

2019–2020

Academic Projects

Gas-Detect Board

Designed a 2-layer PCB for real-time gas leak detection, implementing a 3-level alert system with TFT display, buzzer alarm, and automatic fan activation, validated through perfboard prototyping.

Brushless Motor Controller

Developed a high-efficiency 2-layer PCB controller for BLDC motors in electric vehicles, integrating MOSFETs and IGBTs with optimized thermal management for stable operation under load.

DC Motor Speed Controller

Built a 2-layer PCB implementing PWM-based speed control for DC motors, centered on a 555 timer with MOSFET switching and a freewheeling diode for circuit protection.

Weather Station

Conceived a 2-layer PCB for real-time environmental monitoring, combining an STM32 MCU, Wi-Fi connectivity, and DHT11 and rain sensors for continuous data acquisition.

Skills

Hardware Design: Altium Designer, HyperLynx (Signal & Power Integrity), Saturn PCB Toolkit, PSpice for TI, LTspice, Proteus.

Programming: C, C++, Python, VHDL, MATLAB, LabVIEW.

Protocols: UART, SPI, I²C, CAN, SDMMC, USB, I²S, Ethernet, UIM.

Soft Skills: Innovation & Creativity, Adaptability, Problem-solving, Teamwork.

Certificate & Volunteer Experience

Certificate: Altium Designer Pro.

IEEE SSCS Chapter – Active Member & Trainer

2022–2025

Led training workshops in PCB design, embedded systems, and circuit prototyping; developed educational materials and mentored students and young engineers.

Languages

• **Arabic:** Native

• **English:** Upper-Intermediate

• **French:** Upper-Intermediate